Data Flow Diagram

Date	15 JULY 2026
Project Name	Smart Lender: AI-Powered Loan Approval Prediction System
Maximum Marks	2 Marks

Data Flow Diagram

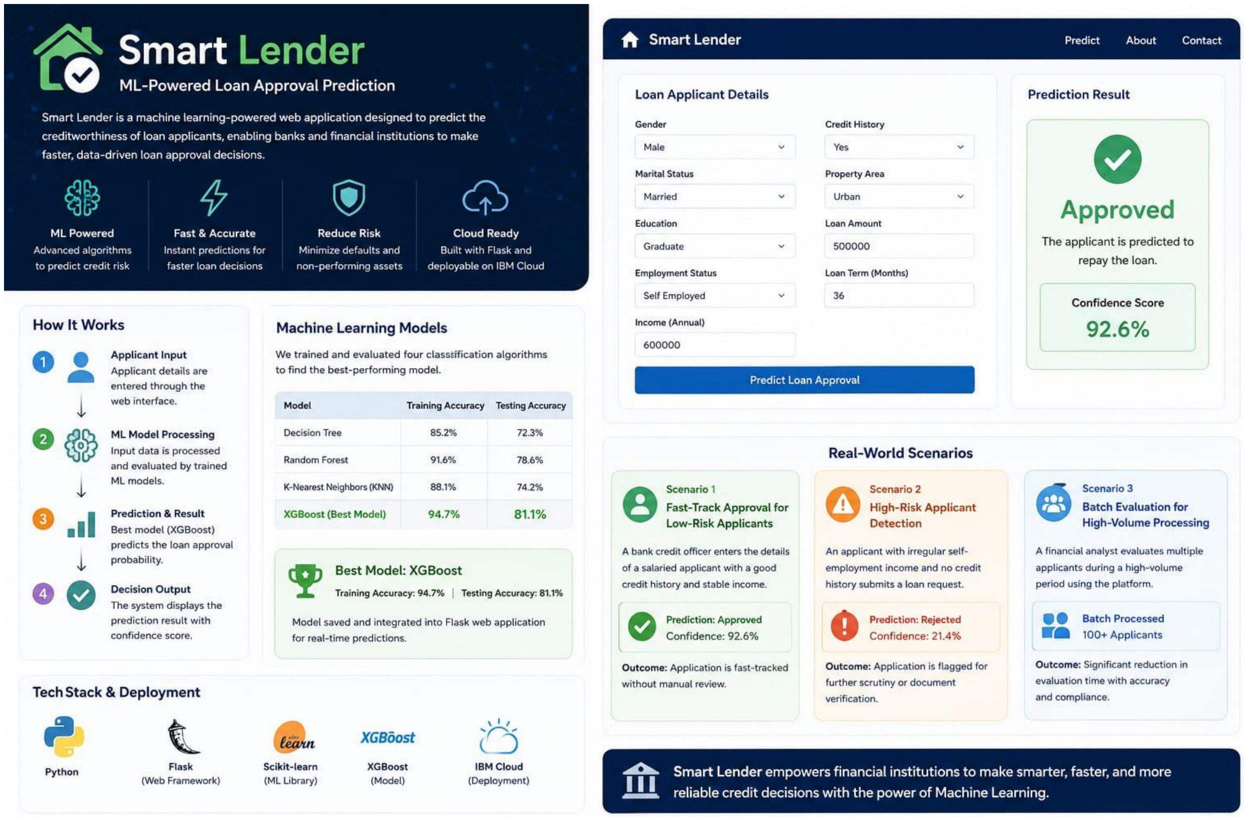
Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.


Smart Lender
ML-Powered Loan Approval Prediction
Smart Lender is a machine learning-powered web application designed to predict the creditworthiness of loan applicants, enabling banks and financial institutions to make faster, data-driven loan approval decisions.
ML Powered
Advanced algorithms to predict credit risk
Fast & Accurate
Instant predictions for faster loan decisions
Reduce Risk
Minimize defaults and non-performing assets
Cloud Ready
Built with Flask and deployable on IBM Cloud
How It Works
1 Applicant Input
Applicant details are entered through the web interface.
2 ML Model Processing
Input data is processed and evaluated by trained ML models.
3 Prediction & Result
Best model (XGBoost) predicts the loan approval probability.
4 Decision Output
The system displays the prediction result with confidence score.
Machine Learning Models
We trained and evaluated four classification algorithms to find the best-performing model.
Model | Training Accuracy | Testing Accuracy
Decision Tree | 85.2% | 72.3%
Random Forest | 91.6% | 78.6%
K-Nearest Neighbors (KNN) | 88.1% | 74.2%
XGBoost (Best Model) | 94.7% | 81.1%
Best Model: XGBoost
Training Accuracy: 94.7% | Testing Accuracy: 81.1%
Model saved and integrated into Flask web application for real-time predictions.
Tech Stack & Deployment
Python
Flask (Web Framework)
Scikit-learn (ML Library)
XGBoost (Model)
IBM Cloud (Deployment)
Smart Lender
Predict
About
Contact
Loan Applicant Details
Gender
Male
Credit History
Yes
Marital Status
Married
Property Area
Urban
Education
Graduate
Loan Amount
500000
Employment Status
Self Employed
Loan Term (Months)
36
Income (Annual)
600000
Predict Loan Approval
Prediction Result
Approved
The applicant is predicted to repay the loan.
Confidence Score
92.6%
Real-World Scenarios
Scenario 1
Fast-Track Approval for Low-Risk Applicants
A bank credit officer enters the details of a salaried applicant with a good credit history and stable income.
Prediction: Approved
Confidence: 92.6%
Outcome: Application is fast-tracked without manual review.
Scenario 2
High-Risk Applicant Detection
An applicant with irregular self-employment income and no credit history submits a loan request.
Prediction: Rejected
Confidence: 21.4%
Outcome: Application is flagged for further scrutiny or document verification.
Scenario 3
Batch Evaluation for High-Volume Processing
A financial analyst evaluates multiple applicants during a high-volume period using the platform.
Batch Processed
100+ Applicants
Outcome: Significant reduction in evaluation time with accuracy and compliance.
Smart Lender empowers financial institutions to make smarter, faster, and more reliable credit decisions with the power of Machine Learning.